

APTITUDE TEST (SET-A)

Instructions:

- Each question carries 1 mark
- All questions are compulsory

Section 1: Logical Reasoning (10 Marks)

1. Find the next number in the series:

4, 6, 9, 13, 18, ____

- A) 23
- B) 24
- C) 25
- D) 26

2. Find the missing term:

3, 12, 48, 192, ____

- A) 576
- B) 720
- C) 768
- D) 800

3. In a certain coding system, each letter is replaced by its position value multiplied by 2, What is the code for "BAD"?

- A) 2-2-8
- B) 4-2-8
- C) 4-1-8
- D) 2-4-8

4. Find the odd one out:

64, 125, 216, 343, 512, 729, 800

- A) 125
- B) 343
- C) 800
- D) 729

5. Statements:

All engineers are graduates.

Some graduates are unemployed.

Conclusions:

I. Some engineers are unemployed

II. Some graduates are engineers

- A) Only I follows
- B) Only II follows
- C) Both follow
- D) None follow

6. Find the next term:

AB, DE, HI, MN, ____

- A) QR
- B) RS
- C) ST
- D) TU

Section 2: Basic Mathematics(10 Marks)

7. If in a code language, PEN = 123 and INK = 456, then PIN = ?

- A) 145
- B) 132
- C) 153
- D) 142

8. A is the sister of B. B is the father of C. C is the son of D. How is A related to D?

- A) Sister
- B) Mother
- C) Sister-in-law
- D) Aunt

9. What is the angle between the hour and minute hand at 5:20?

- A) 40°
- B) 50°
- C) 60°
- D) 70°

10. If today is Friday, what day will it be after 75 days?

- A) Sunday
- B) Monday
- C) Tuesday
- D) Wednesday

11. If $X + 1/X = 4$, find $X^2 + 1/X^2 = ?$

- A) 12
- B) 14
- C) 16
- D) 10

12. Solve the quadratic equation:

$$x^2 - 7x + 10 = 0$$

- A) $x = 2, 5$
- B) $x = 3, 4$
- C) $x = 1, 10$
- D) $x = 5, 6$

13. A train crosses a pole in 8 seconds and a platform of 160 m in 24 seconds. Find the speed of the train.

- A) 20 m/s
- B) 25 m/s
- C) 30 m/s
- D) 35 m/s

14. A sum becomes ₹1210 in 2 years at 10% compound interest. Find the principal.

- A) 1000
- B) 1100
- C) 900
- D) 950

15. A and B can do a work in 15 days. B alone can do it in 20 days. In how many days can A alone do it?

- A) 40
- B) 60
- C) 30
- D) 25

16. Find the remainder when 3^{50} is divided by 4.

- A) 1
- B) 2
- C) 3
- D) 0

17. If $\sin \theta = 5/13$, find $\cos \theta$.

- A) $12/13$
- B) $13/5$
- C) $5/12$
- D) $12/5$

18. A person travels equal distances at speeds of 20 km/h and 30 km/h. Find the average speed.

- A) 24 km/h
- B) 25 km/h
- C) 26 km/h
- D) 27 km/h

19. How many different arrangements can be made using the letters of the word "LEVEL"?

- A) 60
- B) 120
- C) 30
- D) 20

20. The ratio of two numbers is 3:5 and their sum is 64. Find the larger number.

- A) 24
- B) 30
- C) 40
- D) 50

Section 3: DSA (10 Marks)

21. What is the worst-case time complexity of binary search?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n \log n)$

22. Which data structure is used to implement recursion?

- A) Queue
- B) Stack
- C) Array
- D) Tree

23. What is the average-case time complexity of Quick Sort?

- A) $O(n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(n \log n)$

24. What will be the output of the following code?

Java

```
int i = 3;
```

```
System.out.println(i++ * ++i);
```

- A) 12
- B) 15
- C) 16
- D) 20

25. Which traversal method is used in Breadth First Search?

- A) Stack
- B) Queue
- C) Recursion
- D) Array

26. What is the time complexity of inserting an element into a heap?

- A) $O(1)$
- B) $O(n)$
- C) $O(\log n)$
- D) $O(n \log n)$

27. What will be the output?

Java

```
System.out.println(5 + 5 + "5");
```

- A) 105
- B) 55
- C) 1055
- D) 555

28. What is the best-case time complexity of Bubble Sort?

- A) $O(n)$
- B) $O(n^2)$
- C) $O(\log n)$
- D) $O(1)$

29. What is a hash collision?

- A) Two keys mapped to same index
- B) Array overflow
- C) Stack overflow
- D) Sorting error

30. Which data structure is best for implementing BFS?

- A) Stack
- B) Queue
- C) Tree
- D) Graph